

DEFI QUANTMESH X402

Autonomous Trading Signal & Risk Fusion Router | PS0407 Open Innovation Proposal

SELECTED TRACK

PS0407 (Open Innovation)

CORE TECH PRIMITIVE

Algorand x402 + 4-Tx Atomic Group

TEAM STRUCTURE

4 Members | Modular API-First

1. Executive Summary & PS0407 Framing

DeFi QuantMesh x402 is an autonomous, pay-per-signal quantitative trading and risk aggregation mesh submitted under **PS0407 (Open Innovation)**. It solves a major pain point for crypto traders and automated sniper bots: accessing multi-source quantitative market intelligence without paying thousands of dollars in monthly SaaS subscriptions or risking un-orchestrated API execution failures.

Why PS0407 (Open Innovation) is the Winning Framing

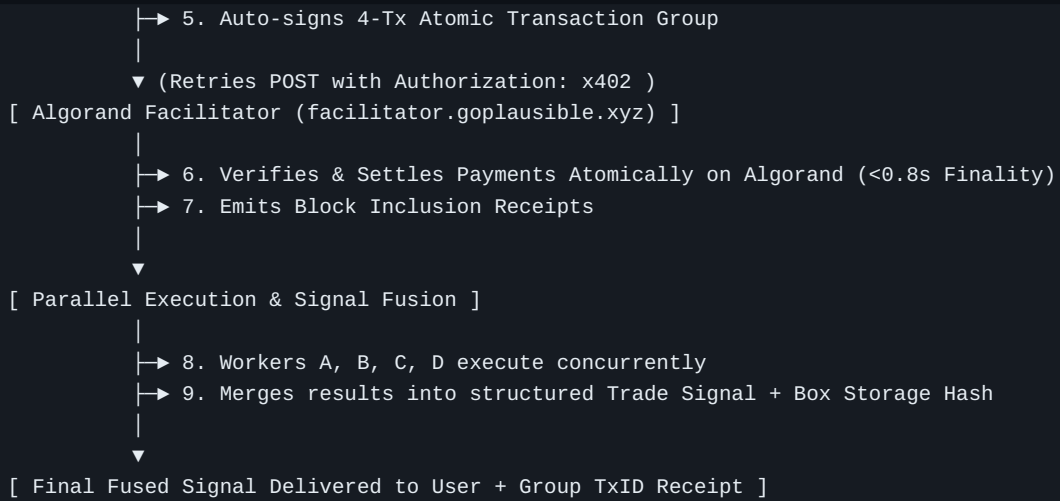
While PS0404 restricts projects to generic multi-agent routing, **PS0407 allows us to build a domain-specific financial product** (Social Sentiment + On-Chain Whale Metrics + Technical Analysis + Risk-Weighted Signal Fusion) while **100% satisfying all mandatory Agentic Payment (x402) rules** outlined in the hackathon prompt poster.

Strict Mandatory Image Guidelines Compliance Checklist

- **x402 Payment Challenge (402 Payment Required):** Every sub-agent server responds with HTTP status 402 and x402 headers defining required micro-pricing (\$0.001 - \$0.002 USDC ASA).
- **Client Signs & Retries Automatically:** The client uses `@x402/fetch` / `@x402/avm` to sign the multi-output atomic transaction group and retries seamlessly.
- **Facilitator Verifies & Settles Payment:** Integrates `https://facilitator.goplausible.xyz` to verify and settle payouts on Algorand Testnet.
- **Transaction-Linked Receipt:** Aggregated trade signal response includes the Algorand Group Transaction ID (`txId`) and logs output hashes in Box Storage / Transaction Notes for auditability.
- **Pay-Per-Use Model:** Eliminates \$2,000/month Bloomberg/Glassnode SaaS subscriptions in favor of \$0.006 USDC per execution.

2. System Architecture & 4-Agent Mesh Workflow

```
[ Client / Automated Bot Requests Signal ]
  |
  | 1. POST /api/v1/orchestrate { tokenSymbol: "ALGO", timeframe: "1h" }
  |
  ▼
[ DeFi QuantMesh Router Engine (@x402/hono) ]
  |
  |→ 2. Task Decomposition across 4 Specialized Worker Nodes:
  |   • Worker A: Social Sentiment Agent (FinBERT) ($0.0020 USDC)
  |   • Worker B: On-Chain Whale Metrics Agent ($0.0020 USDC)
  |   • Worker C: TA Indicator Engine (EMA/RSI/MACD) ($0.0010 USDC)
  |   • Worker D: Risk-Weighted Fusion Engine ($0.0010 USDC)
  |
  |→ 3. Calculates Total Micro-Cost: $0.0060 USDC ASA
  |→ 4. Assembles Algorand Atomic Group (4 Payout Tx + 1 Router Tx)
  |
  ▼ (Returns HTTP 402 Payment Required + Atomic Group Spec)
[ Client Wallet SDK (@x402/avm) ]
  |
```



3. Technical Stack & 4-Member Work Division Plan

LAYER / SUBFOLDER	ASSIGNED MEMBER	TECH STACK SELECTED	CORE RESPONSIBILITIES
/frontend	Member 1 (UI & Pitch Lead)	Next.js, TailwindCSS, ShadCN UI, @x402/fetch	Build trading dashboard, token query input, live x402 flow visualizer, display AlgoExplorer receipt link, lead 5-min demo pitch.
/orchestrator	Member 2 (Blockchain Lead)	Node.js, TypeScript, @x402/hono , algosdk	Build router endpoint, formulate 4-tx Algorand Atomic Group, integrate Plausible facilitator, handle Box Storage logging.
/agent-sentiment & /agent-fusion	Member 3 (AI Worker Engineer 1)	FastAPI / Hono, HuggingFace FinBERT API, Gemini 1.5 Flash	Build Agent A (Social Sentiment NLP) and Agent D (Risk-Weighted Signal Fusion) microservices wrapped in x402 middleware.
/agent-onchain & /agent-ta	Member 4 (AI Worker Engineer 2)	FastAPI / Hono, Algorand Indexer API, CoinGecko, TA-Lib	Build Agent B (On-Chain Whale Metrics) and Agent C (TA Technical Indicator Engine) microservices wrapped in x402 middleware.

4. APIs Required & Zero-Cost Financial Breakdown

Required APIs & Credentials

- **Algorand Testnet Node / Indexer:** AlgoNode Public RPC (<https://testnet-api.algonode.cloud>) — **\$0.00 (Free Public)**
- **x402 Facilitator Endpoint:** Plausible Facilitator (<https://facilitator.goplausible.xyz>) — **\$0.00 (Free Hackathon Tier)**
- **NLP Sentiment Model:** Hugging Face Inference API ([ProsusAI/finbert](#)) — **\$0.00 (Free Tier)**
- **Market OHLCV & Price Data:** CoinGecko v3 API / DeFi Llama API — **\$0.00 (Free Public Tiers)**
- **Signal Fusion & Reasoning:** Google Gemini 1.5 Flash / Groq Llama-3-70B — **\$0.00 (Free Tier)**

Production Micro-Transaction Unit Economics

COMPONENT	PRICE PER CALL (USDC)	DESTINATION
Agent A: Social Sentiment Agent (FinBERT)	\$0.0020 USDC	Worker Agent A Wallet
Agent B: On-Chain Whale Metrics Agent	\$0.0020 USDC	Worker Agent B Wallet
Agent C: TA Technical Indicator Engine	\$0.0010 USDC	Worker Agent C Wallet
Agent D: Risk-Weighted Fusion Engine	\$0.0010 USDC	Worker Agent D Wallet
Algorand Blockchain Network Fees (5 TxS)	\$0.0005 USDC (0.005 ALGO)	Algorand Validators
QuantMesh Router Protocol Markup	\$0.0005 USDC	QuantMesh Treasury Wallet
Total Charged Per Fused Trading Signal	\$0.0070 USDC	Single Algorand Atomic Group Settlement

5. Real-World Use Case & Existing Product Limitations (Hinglish Case Study)

This section provides a practical real-world scenario demonstrating why traditional LLMs fail and how DeFi QuantMesh x402 delivers an unbeatable solution for judges.

Real-World Industry Scenario: "The High-Volatility Breakout & Risk Signal Request"

Scenario: Ek Crypto Trader ya Automated Trading Bot ko ek coin (jaise ALGO ya SOL) mein position leni hai. Wo chahta hai ki minute-level par usko Sentiment, On-Chain Whale Flow, aur Technical Indicators (EMA/RSI/MACD) teeno ka ek composite aggregated signal mile, taaki wo fakeouts se bach sake.

Existing Products & ChatGPT / Claude Ki Limitations (Kyun Fail Hote Hain):

- 1. The Subscription Hell & Fragmented Tools:** Sentiment ke liye Santiment ka \$199/month, On-Chain ke liye Glassnode ka \$799/month, aur Charting ke liye TradingView ka \$50/month subscription lena padta hai. Agar aap mahine mein sirf 5 baar trade karte ho, toh bhi aapko \$1,000+ per month fixed charge dena padta hai.
- 2. ChatGPT Has No Live Data or Wallet Access:** ChatGPT real-time market orderbook, live indexer data, ya sentiment models ko natively query nahi kar sakta. Aur sabse badi baat—ChatGPT ke paas crypto wallet nahi hai, toh wo kisi paid live data API ko \$0.002 USDC instant pay karke real-time signals fetch nahi kar sakta.
- 3. Un-orchestrated API Execution Failures:** Trading bot jab multiple Web2 APIs call karta hai, aur 3rd API (On-Chain metrics) crash ya timeout ho jaati hai, toh pehle 2 APIs paise charge kar leti hain par bot ko partial/corrupted data milta hai, jisse trading loss ho sakta hai.

DeFi QuantMesh x402 Kaise Solve Karta Hai (How We Win):

- 1. Pay-Per-Signal Micropayments:** No monthly subscriptions! User ya Trading Bot sirf tab pay karta hai jab wo signal request karta hai (\$0.007 USDC per signal).
- 2. Algorand Native Atomic Transfers (The Secret Weapon):** Router sabhi 4 worker nodes (Sentiment, On-Chain, TA Engine, Fusion Agent) ke payouts ko ek ****Single Algorand Atomic Group Transaction**** mein bind kar deta hai.
- 3. All-or-Nothing Sub-Second Settlement (<0.8s):** Algorand blockchain par saare sub-agents ko ek saath <0.8 seconds mein payment milti hai. Agar ek bhi agent down hota hai ya response fail karta hai, toh ****transaction instantly rollback ho jata hai aur trader ka 1 paisa bhi nahi kat-ta.****
- 4. On-Chain Tamper-Proof Receipts:** Every signal response header includes an Algorand Group Transaction ID (`txId`) and hash recorded in Algorand Box Storage, giving complete auditability for trading decision logs.

Judge Counter-Punch Cheat Sheet

When Judge Asks: *"Why can't I just ask ChatGPT for trading advice?"*

Your Winning Reply: *"Sir, ChatGPT is a static brain without live blockchain access or a financial wallet. It cannot autonomously query real-time paid indexers, run FinBERT NLP models on live tweets, or execute sub-second micropayments. DeFi QuantMesh is the financial nervous system that orchestrates 4 specialized quantitative nodes and settles payouts risk-free in 0.8 seconds over Algorand x402."*